Week 23 — Extended reading

Shielding, CBFs, Domain Randomisation, Robust RL

STUDY-RL

May 21, 2026

Paper 1 — "Safe Reinforcement Learning via Shielding"

Authors: Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, Ufuk Topcu. **Year:** 2018. **Venue:** AAAI. **arXiv:** 1708.08611.

Summary. Introduces the concept of an RL *shield*: a reactive correct-by-construction component synthesised from an ω-regular safety specification, that overrides the agent's action whenever it would lead to a violation. Both pre-shield (filters action set) and post-shield (corrects chosen action) variants are described. Soundness is proved against the specification.

Why it matters. The canonical reference for the formal-methods side of safe RL. Our `project/shield.py` is a clean implementation of the pre-shield variant for the running $\mathbf{G}\,\neg$danger specification.

Annotated key passage.

> "A correct-by-construction shield is a reactive system that monitors the actions chosen by the learning agent and corrects them if necessary, such that the temporal logic specification is satisfied. The shield is computed offline from the specification and a (possibly abstract) model of the environment."

Commentary. The key conceptual move is the *decoupling* of safety and learning. The shield's soundness does not depend on the learner; the learner's correctness does not depend on the shield. This factoring is what makes the approach scale to deep RL: you can plug a shield in front of any policy.

Paper 2 — "Safe Exploration in Continuous Action Spaces"

Authors: Gal Dalal, Krishnamurthy Dvijotham, Matej Vecerik, Todd Hester, Cosmin Paduraru, Yuval Tassa. **Year:** 2018. **arXiv:** 1801.08757.

Summary. Augments an actor with a final *safety layer* — a closed-form (or differentiable QP) projection that maps the actor's proposed action onto the half-space of safe actions defined by a linearised constraint. The whole pipeline is differentiable so gradients flow through the projection.

Why it matters. Where shielding is the discrete-action poster child, the safety layer is the continuous-action poster child. Closed-form projections for a single linear constraint, OptNet-style QPs for multiple.

Annotated key passage.

> "The Safety Layer is a closed-form linear projection $a^\star = a - [g(s) + \partial g/\partial a \cdot a]_+ / \|\partial g/\partial a\|^2 \cdot \partial g/\partial a$. Crucially, this projection is differentiable, so the safety layer can be inserted at training time without breaking the policy gradient."

Commentary. The differentiability is the whole point. A non-differentiable safety wrapper would force you into surrogate gradient estimators; OptNet (Amos & Kolter 2017) made the QP itself a primitive that PyTorch can backprop through.

Paper 3 — "Domain Randomization for Transferring Deep Neural Networks"

Authors: Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, Pieter Abbeel. **Year:** 2017. **arXiv:** 1703.06907.

Summary. Trains policies on a wide *distribution* of simulator parameters (lighting, friction, textures) and transfers zero-shot to a real robot. The policy must implicitly learn invariances across the distribution.

Why it matters. The simplest robust-RL technique. Often the first thing tried in sim-to-real; frequently *the best* thing tried, because it dovetails with any underlying RL algorithm.

Annotated key passage.

> "The key insight is that, by randomising aspects of the simulator that may otherwise overfit, the network is forced to learn robust features — and the real world simply becomes another sample from the training distribution."

Commentary. The same principle is data augmentation, lifted to RL. The critical knob is the *support* of the randomisation: too narrow and the policy doesn't generalise; too broad and training collapses (no signal). ADR (OpenAI Rubik's Cube, arXiv:1910.07113) automates this expansion.

Paper 4 — "Sim-to-Real Transfer with Foundation Models"

Authors: Longchao Da, Justin Turnau, Thirulogasankar Pranav Kutralingam, Alvaro Velasquez, Paulo Shakarian, Hua Wei. **Year:** 2025. **arXiv:** 2502.13187.

Summary. A survey of how vision–language foundation models are being plugged into the sim-to-real pipeline: as world models, as observation encoders, as reward labellers, and as scene generators for DR.

Why it matters. The 2025 frontier for robust RL. The foundation model gives a near-universal feature space; DR happens *in feature space*, not in pixel space, which is much more sample-efficient.

Annotated key passage.

> "Foundation-model encoders project simulated and real observations into a shared semantic space, transforming the sim-to-real gap from a pixel-statistics problem into a feature-alignment problem. This is the modern descendant of domain randomisation."

Commentary. The clean way to view this: pre-2020 DR was about making the policy invariant; post-2024 DR is about making the encoder invariant. The policy can be much simpler when the encoder already abstracts away nuisance variation.

Paper 5 — "Learning-Based Shielding under Unknown Dynamics"

Authors: (Multi-author, see arXiv). **Year:** 2024. **arXiv:** 2410.07359.

Summary. Extends the shielding framework to settings where the environment model is learned rather than given. Provides probabilistic soundness bounds in terms of model uncertainty (PAC-style).

Why it matters. Lifts the most restrictive assumption of Alshiekh-style shielding (a known one-step model). Critical for real deployments where the world is partially observed and partially learned.

Annotated key passage.

> "We show that, when the learned dynamics model is calibrated, the shield's soundness guarantee degrades gracefully: the probability of an unsafe trajectory is bounded by the model's coverage failure probability plus the spec's reachability margin."

Commentary. This is the bridge from formal methods to statistical learning: shields are no longer "correct or broken" but "correct with probability $1 - \delta$". The same language is showing up in conformal prediction wrappers around RL agents (a 2024 healthcare-RL theme).

Sourced from. `research/safety_rl_sota.md`, sections 2 (shielding, CBFs) and 4 (robust RL, DR).